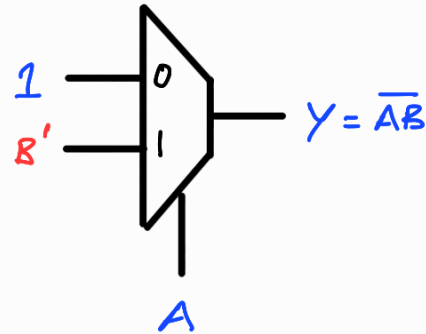


1) Implement the following 2 input logic gates using a single 2x1 Mux, A as select line and B as input

- a) NAND
- b) NOR
- c) XNOR

a) NAND : $\overline{AB}$

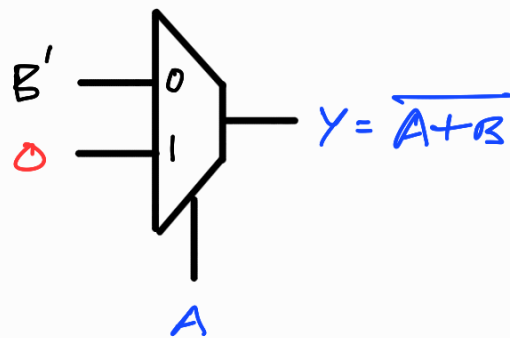
A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	0
1	1	0



When $A = 0$ $Y = \overline{AB} = 1$
 When $A = 1$ $Y = B'$

b) NOR : $\overline{A+B}$

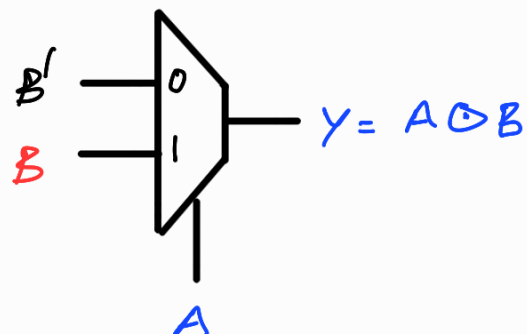
A	B	$\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0



When $A = 0$ $Y = B'$
 When $A = 1$ $Y = 0$

c) XNOR : $A \odot B = \overline{A \oplus B} = AB + \overline{A} \overline{B}$

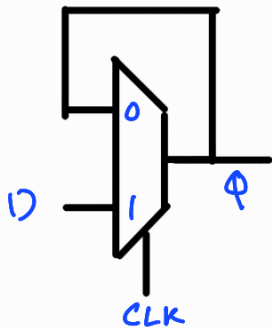
A	B	$AB + \overline{A} \overline{B}$
0	0	1
0	1	0
1	0	0
1	1	1



When $A = 0$ $Y = B'$
 When $A = 1$ $Y = B$

2) Implement D and T Flip flop using 2x1 Mux and logic gates

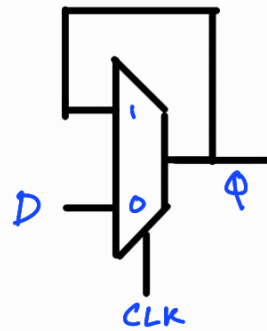
+ve D latch



$$CLK = 0 \Rightarrow Q_{k+1} = 0$$

$$CLK = 1 \Rightarrow Q_{k+1} = D$$

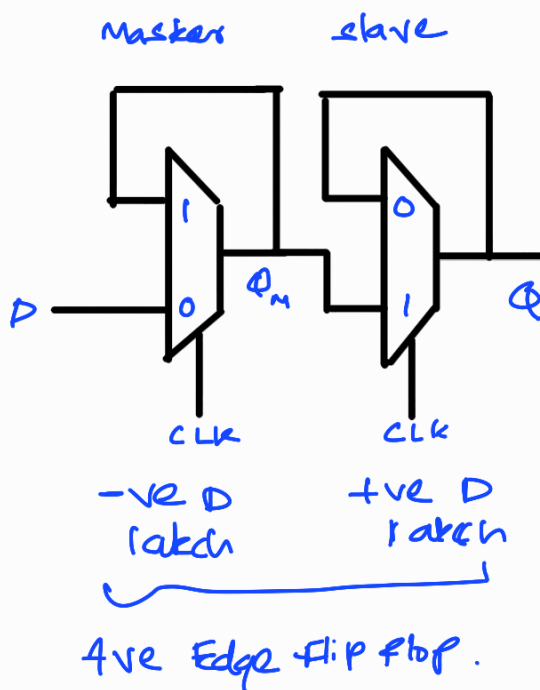
-ve D latch



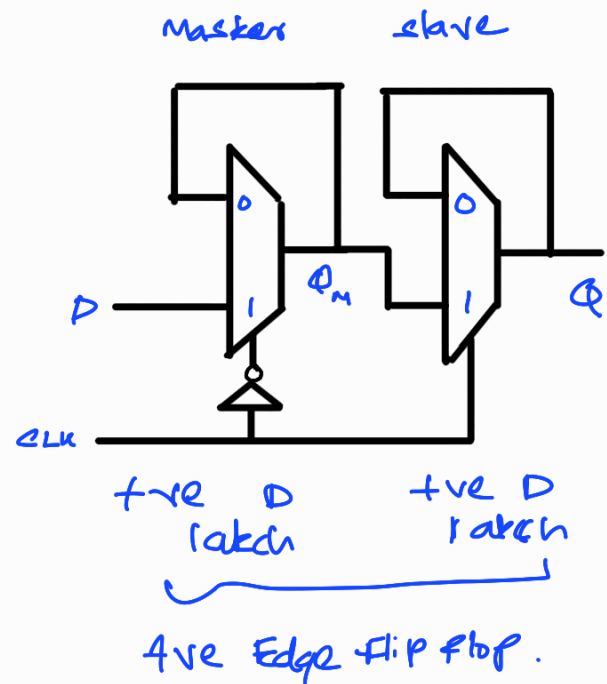
$$CLK = 0 \Rightarrow Q_{k+1} = D$$

$$CLK = 1 \Rightarrow Q_{k+1} = 0$$

Now by using 2 D latches in Master-Slave Configuration

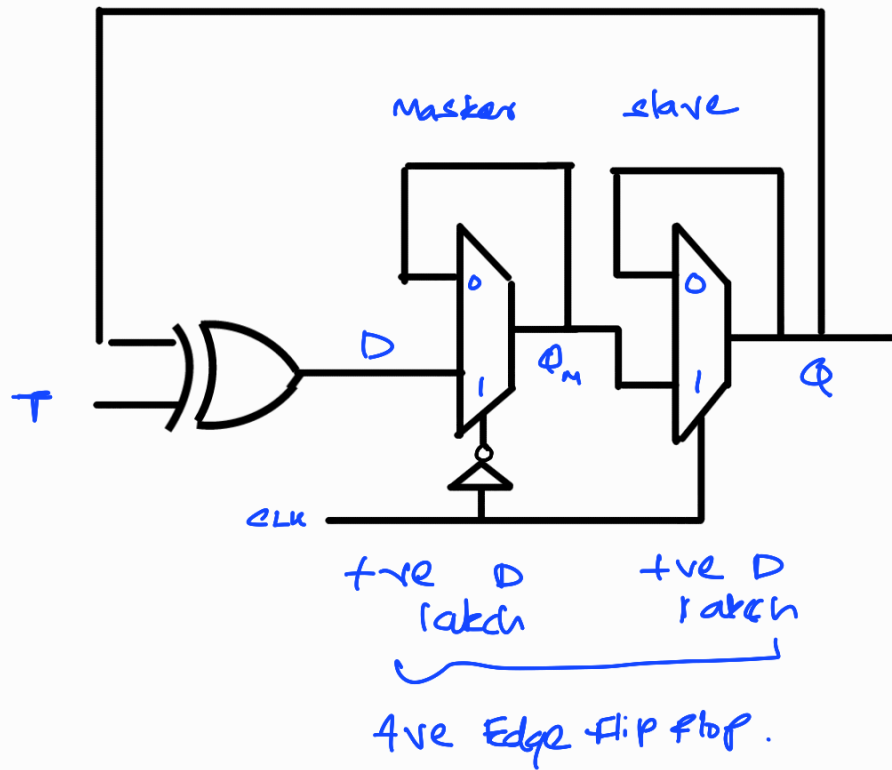


OR

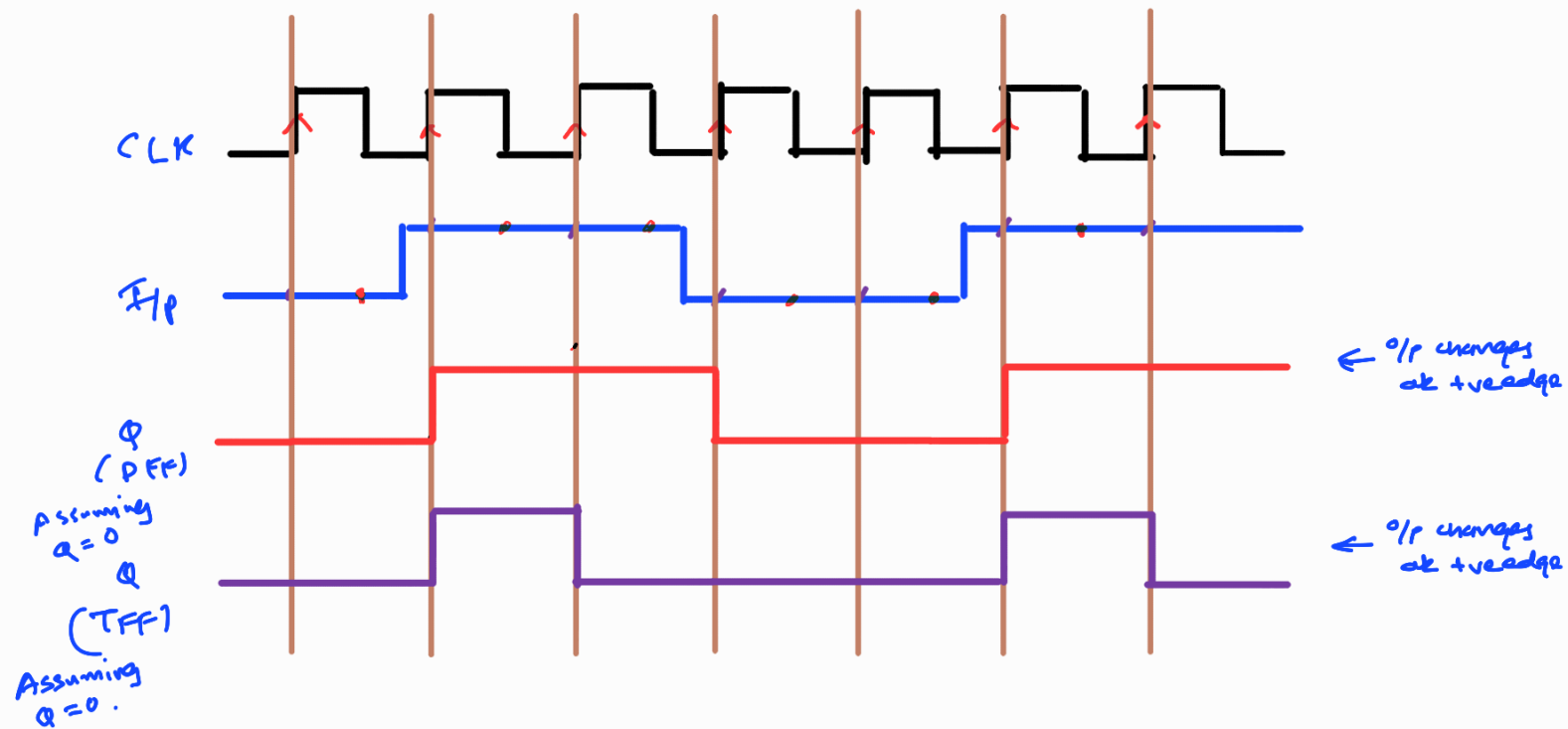


* A positive edge triggered flipflop changes its output at the positive edge of the CLK

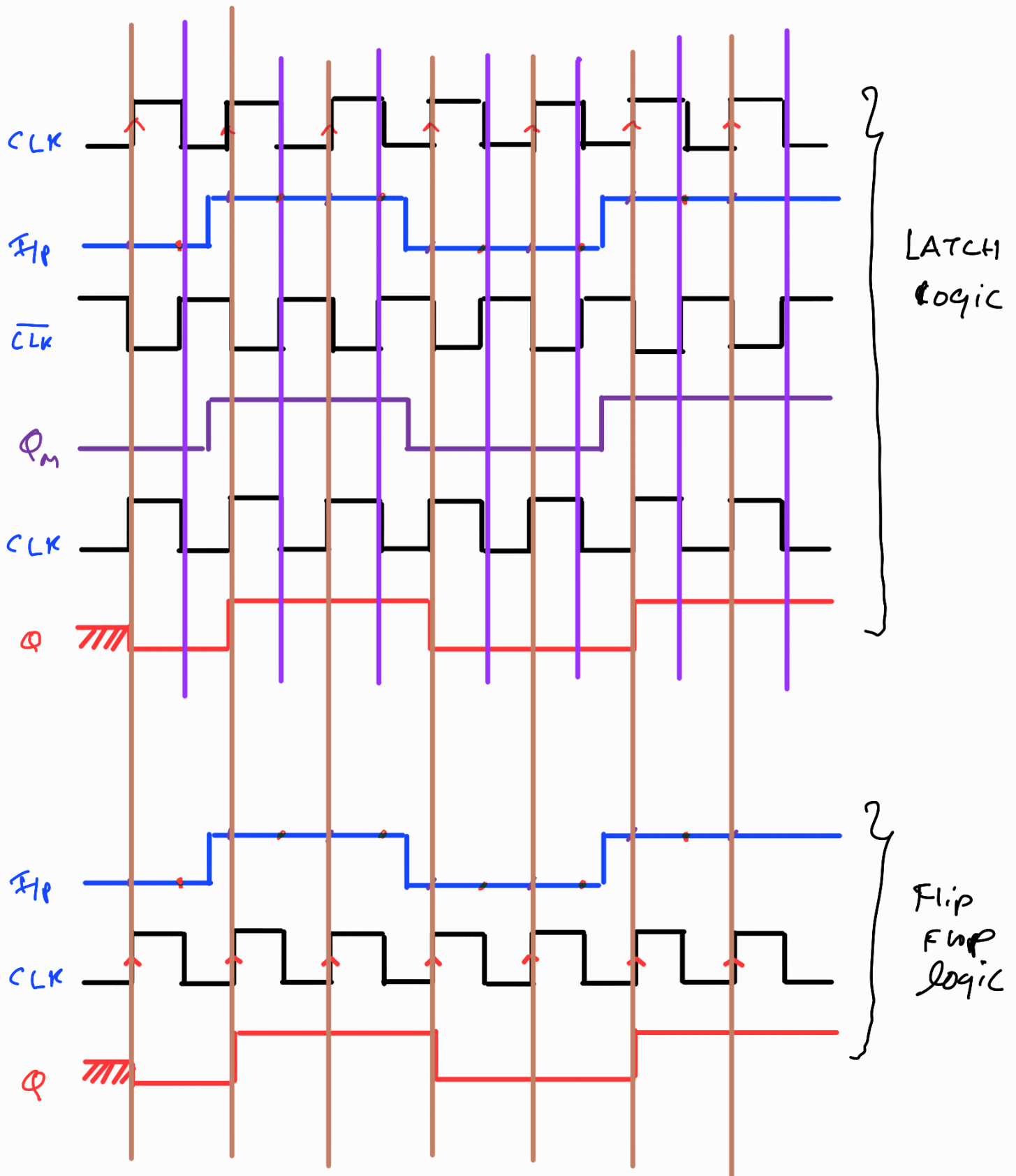
T Flip flop using D flip flop : $D = T \oplus Q_k$



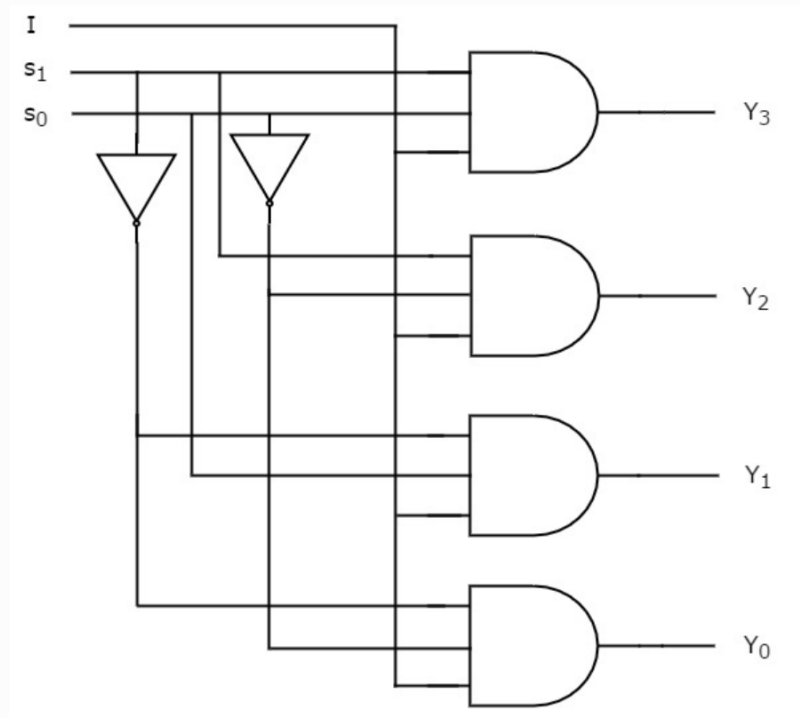
Wave form of a tve edge triggered D & T FF



Verification Using Latch logic & flip flop logic



3) Construct the truth table and output expressions Y_0, Y_1, Y_2 and Y_3 for the combinational circuit provided below, with input I and S_1 (MSB), S_0 as select lines. comment on its operation by comparing it with a multiplexer.



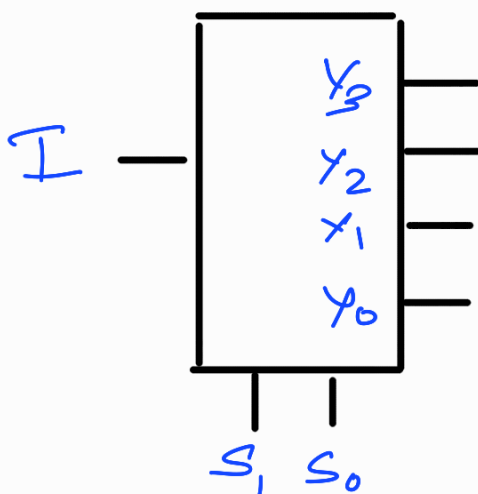
$$Y_3 = I S_1 S_0$$

$$Y_2 = I S_1 S_0'$$

$$Y_1 = I S_1' S_0$$

$$Y_0 = I S_1' S_0'$$

S1	S0	Y3	Y2	Y1	Y0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0



is a 1x4 Demultiplexer.
which routes single
input into 4 output
lines based on select
lines.

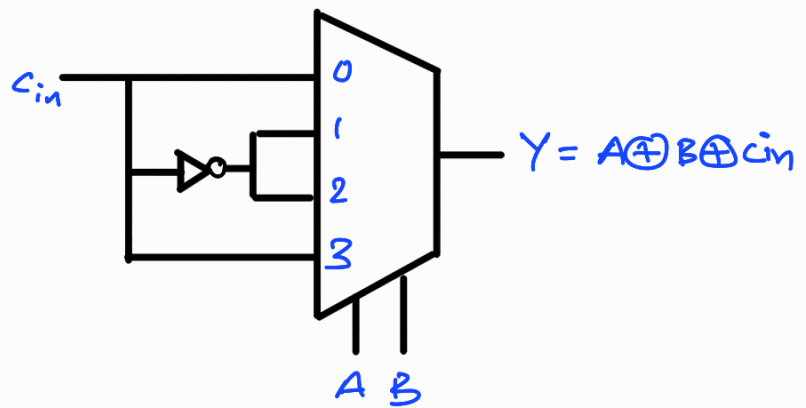
4) A 4:1 multiplexer is to be used for generating the output carry of a full adder. A and B are the bits to be added while C_{in} is the input carry and C_{out} is the output carry. A and B are to be used as the select bits with A being the more significant select bit. Assign values to the inputs of the multiplexer such that the output of the multiplexer is equal to the sum of full adder.

The expression of sum in full adder is

$$Y = A \oplus B \oplus C_{in}$$

Taking A, B as select lines and C_{in} as input

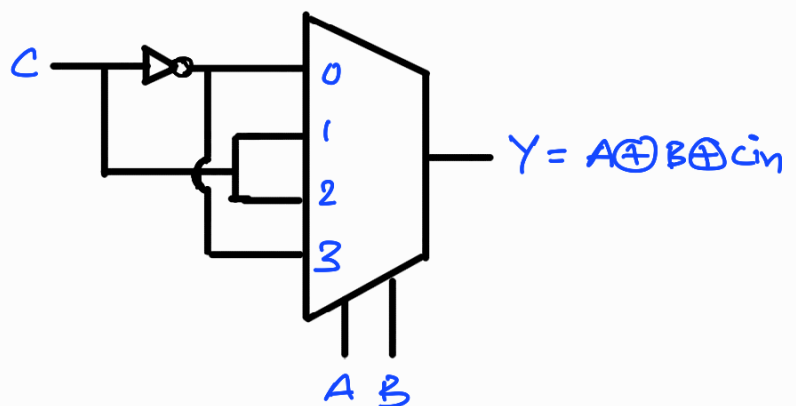
A	B	C_{in}	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



5) Design a combinational circuit with three inputs, A, B and C and one output Y, using 4 × 1 MUX and a minimum number of 2-input logic gates. The circuit should output a 1 if the number of 1's in the input is even and 0 otherwise.

Taking A, B as select lines and C as input

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0



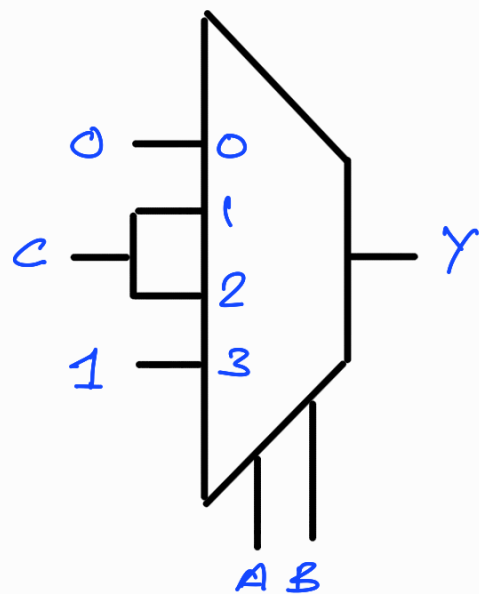
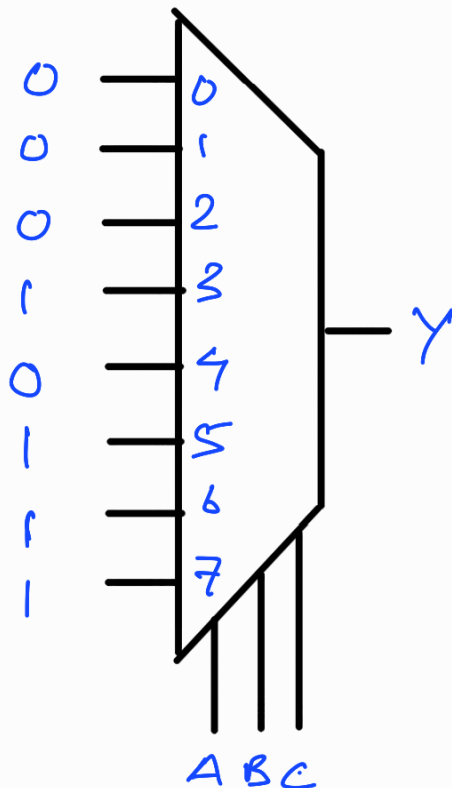
6) Design a 3-bit combinational circuit for which the output is 1 when the number of 1's is greater than 0's and 0 otherwise. This circuit can also be called a majority circuit. Design the circuit using 8x1 Mux and 4x1 Mux . consider A(MSB), B, C as inputs

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

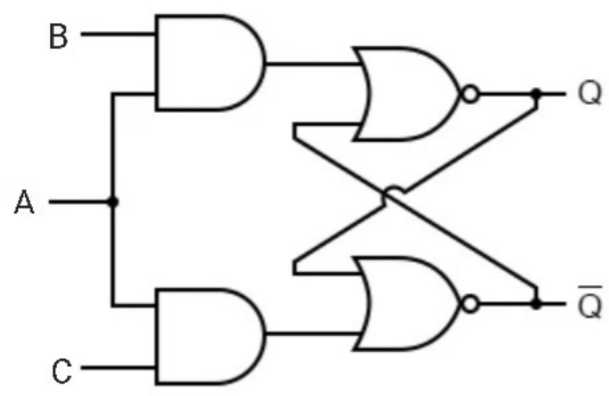
AB	D ₀	D ₁	D ₂	D ₃
00	0	01	10	11
01	0	2	4	6
10	1	3	5	7
11	0	C	C	1

using 8x1 MUX:

using 4x1 Mux:



7) Analyze the sequential circuit provided below, construct its truth table and comment on its operation



• From Boolean Algebra

① $0 \cdot A = 0 \rightarrow$ 0 i/p to AND
Take forces
o/p as 0

② $\overline{1+A} = 0 \rightarrow$ 1 i/p to NOR
Take forces
o/p as 1

A=0

$Q^+ = Q$
 $\overline{Q}^+ = \overline{Q}$
Same state

A=1

$Q^+ = Q$
 $\overline{Q}^+ = \overline{Q}$

A=1

$Q^+ = 1$
 $\overline{Q}^+ = 0$

A=1

$Q^+ = 0$
 $\overline{Q}^+ = 1$

A=1

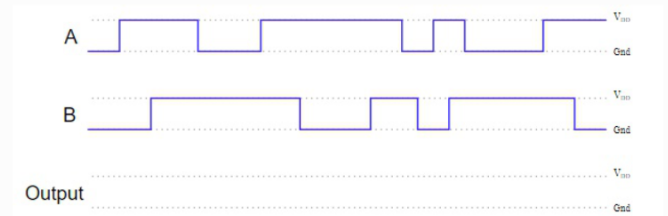
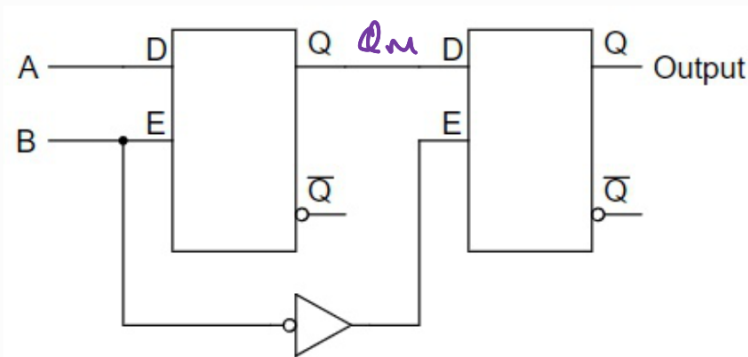
$Q^+ = 0$
 $\overline{Q}^+ = 0$

A	B	C	Q^+	$\overline{Q}^+$	State
0	X	X	Q	$\overline{Q}$	Previous state
1	0	0	Q	$\overline{Q}$	Previous state
1	0	1	1	0	Set
1	1	0	0	1	Reset
1	1	1	0	0	Invalid.

The provided circuit is an gated SR Latch

8) Determine the final output states over time for the following circuit, built from D-type gated latches:

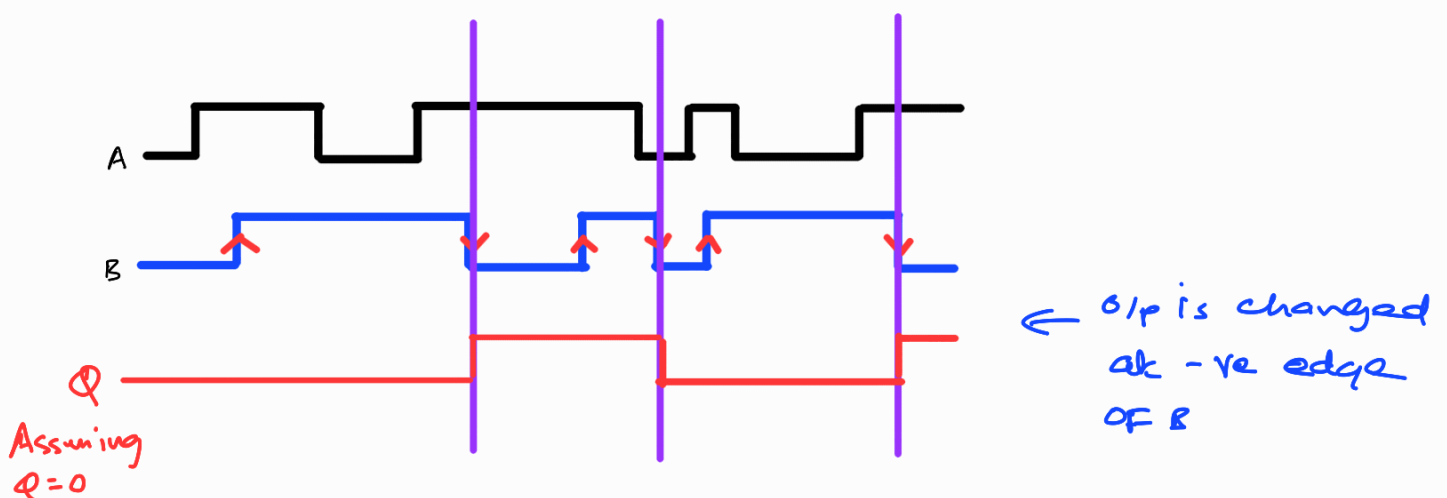
At what specific times in the pulse diagram does the final output assume the input's state?
How does this behavior differ from the normal response of a D-type latch?



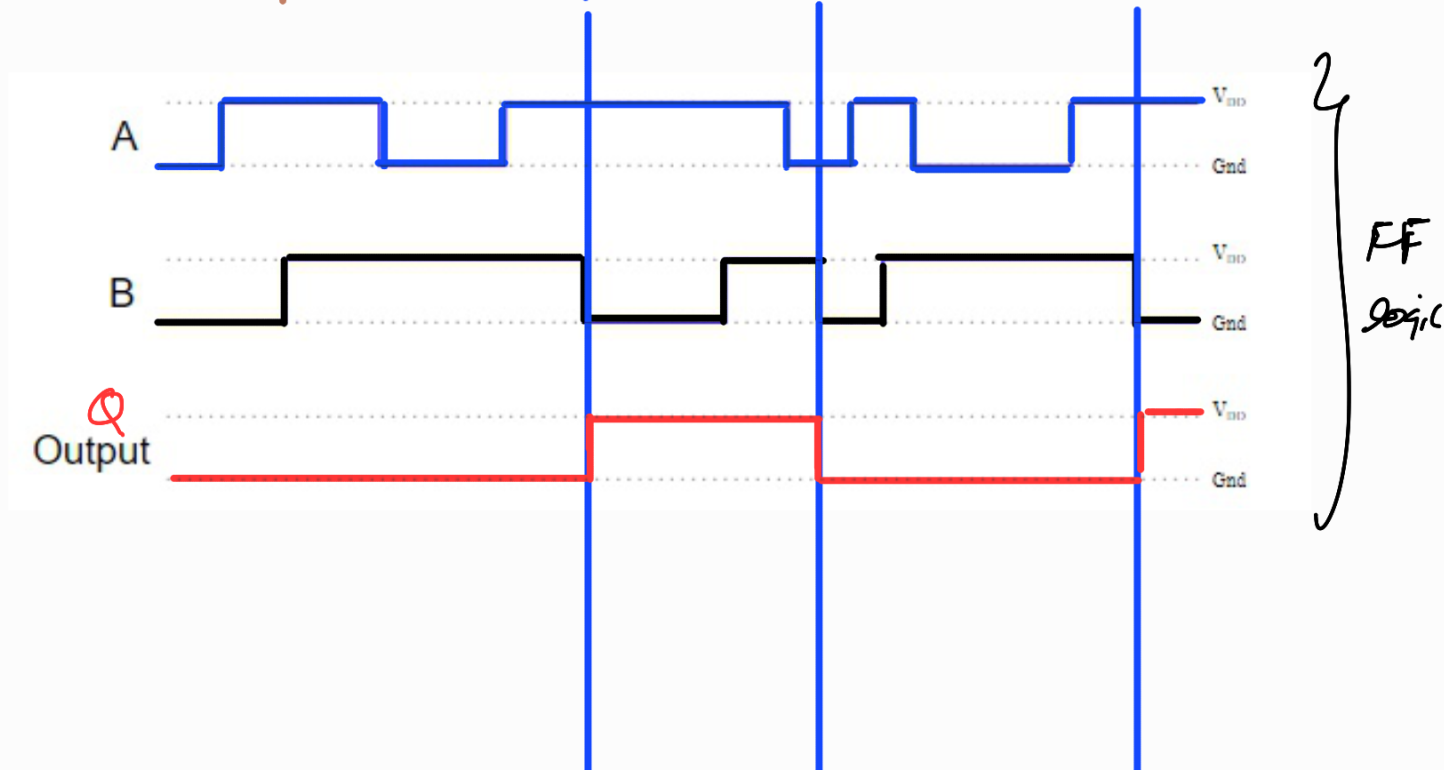
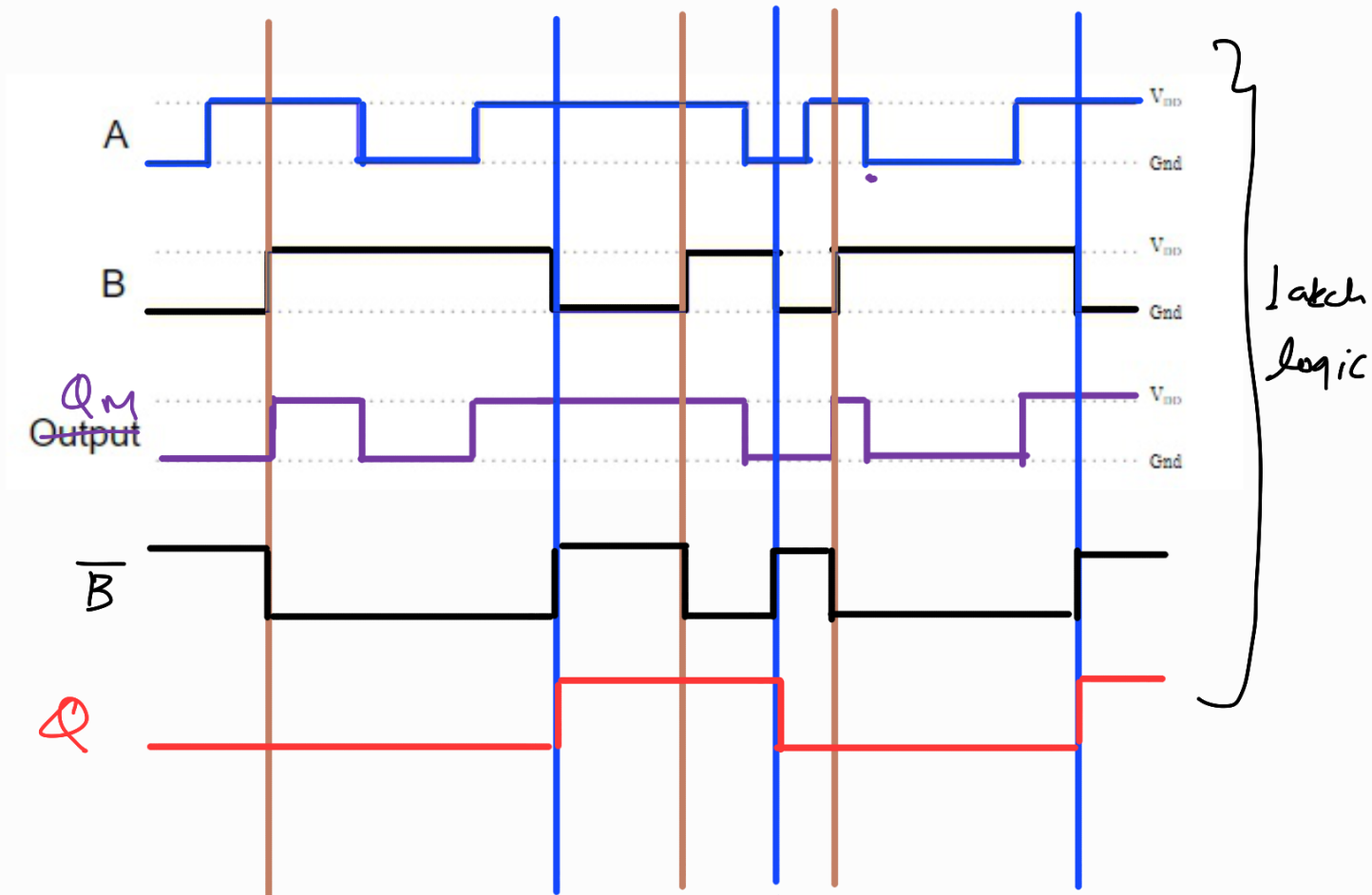
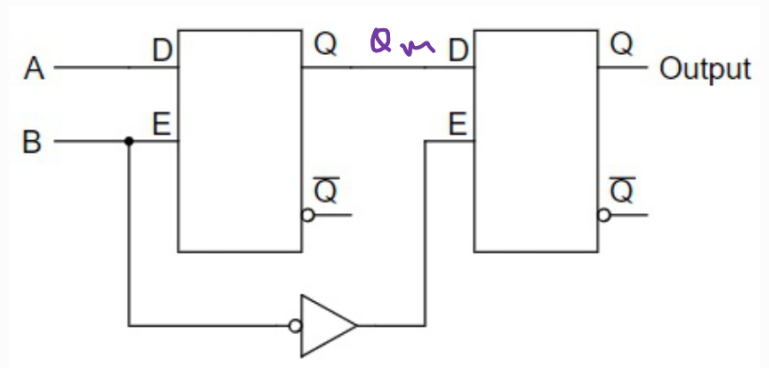
- Circuit provided is negative edge sensitive (0)

A negative edge triggered flip flop changes its output at the negative edge of the clock

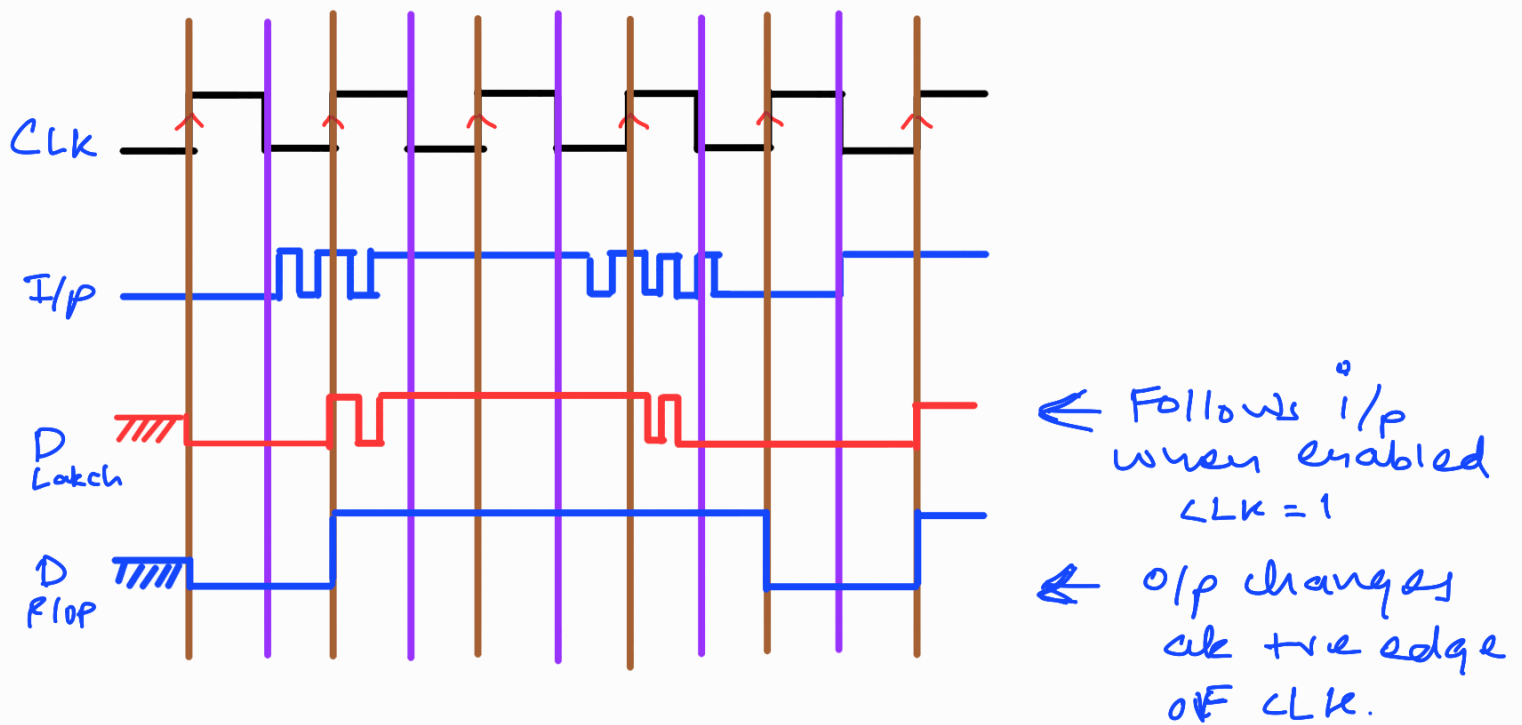
- Normal D latch are level sensity (Q)



Verification using Latch logic and flip flop logic

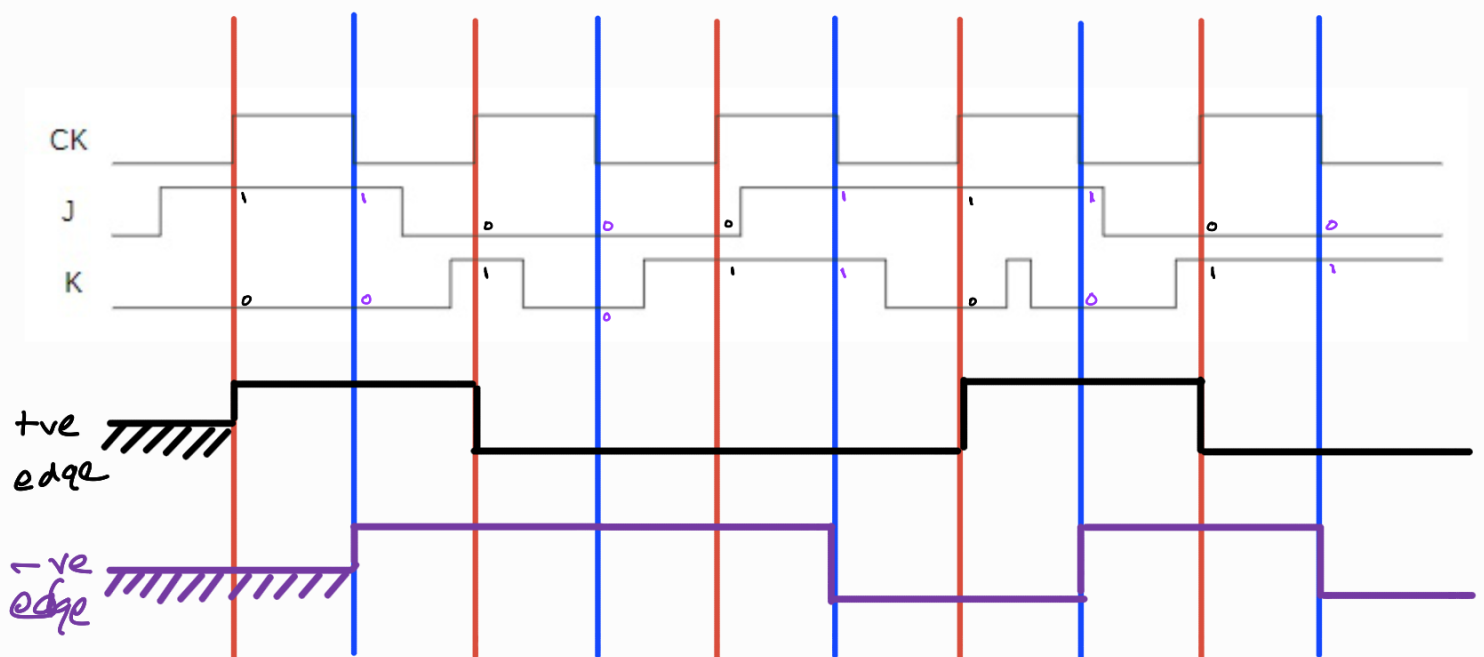


9) Trace the outputs of a D latch and D Flip Flop (positive edge triggered) for the input and clock signals provided below. Compare the outputs and comment on the observations.



10) For the input sequence in the figure, draw the output waveform of:

- a negative edge-triggered JK flip-flop, → o/p @ -ve edge update
- a positive edge-triggered JK flip-flop. → o/p @ +ve edge update

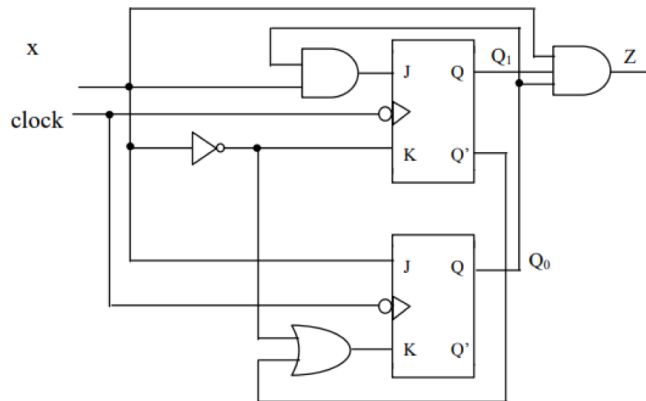


Tutorial-5

(Q11 to Q20)

11) Analyze the synchronous sequential circuit in the figure using the following state assignment:

$Q_1 Q_0$
 A: 0 0
 B: 0 1
 C: 1 1
 D: 1 0



Qdn:

$$J_1 = x Q_0$$

$$K_1 = \overline{x}$$

$$J_0 = x$$

$$K_0 = x + \overline{Q_1}$$

$$Z = x Q_1 Q_0$$

	Q_1	Q_0	x	J_1	K_1	J_0	K_0	Q_1^+	Q_0^+	Z
A:	0	0	0	0	1	0	1	0	0	0
	0	0	1	0	0	1	1	0	1	0
B:	0	1	0	0	1	0	1	0	0	0
	0	1	1	1	0	1	1	1	0	0
C:	1	0	0	0	1	0	0	0	0	0
	1	0	1	0	0	1	1	1	1	0
D:	1	1	0	0	1	0	0	0	1	0
	1	1	1	1	0	1	1	1	0	0

12) Given below are the excitation and output functions of a Moore model synchronous sequential circuit.

$$J_1 = (xQ_0)'\quad K_1 = x + Q_0\quad T_0 = xQ_1 + x'Q_0\quad Z = Q_1Q_0'$$

Analyze the synchronous sequential circuit.

Soln:

$$J_1 = \overline{(xQ_0)} \quad K_1 = x + Q_0 \quad T_0 = xQ_1 + \overline{x}Q_0 \quad Z = Q_1\overline{Q_0}$$

Q_1Q_0	x	J_1	K_1	T_0	Z
00	0	1	0	0	0
00	1	0	1	0	0
01	0	1	1	1	0
01	1	1	1	0	0
10	0	1	0	0	1
10	1	0	1	1	1
11	0	1	1	1	0
11	1	1	1	1	0

13) Given below are the state table and state assignment for a synchronous sequential circuit. Realize the circuit using the D flip-flops.

State Assignment $Q_2Q_1Q_0$	Present State	Next state, Output	
		$X = 0$	$X = 1$
0 0 0	A	B,0	E,0
0 0 1	B	A,1	C,1
0 1 0	C	B,0	C,1
0 1 1	D	C,0	E,0
1 0 0	E	D,1	A,0

Soln:

x	Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	D_2	D_1	D_0	y
0	0	0	0	0	0	1	0	0	1	0
0	0	0	1	0	0	0	0	0	0	1
0	0	1	0	0	0	1	0	0	1	0
0	0	1	1	0	1	0	0	1	0	0
0	1	0	0	0	1	1	0	1	1	1
0	1	0	1	x	x	x	x	x	x	x
0	1	1	0	x	x	x	x	x	x	x
0	1	1	1	x	x	x	x	x	x	x
1	0	0	0	1	0	0	1	0	0	0
1	0	0	1	0	1	0	0	1	0	1
1	0	1	0	0	1	0	0	1	0	1
1	0	1	1	1	0	0	1	0	0	0
1	1	0	0	0	0	0	0	0	0	0
1	1	0	1	x	x	x	x	x	x	x
1	1	1	0	x	x	x	x	x	x	x
1	1	1	1	x	x	x	x	x	x	x

D_2

xQ_2	Q_1Q_0	00	01	11	10
00	0 ₀	0 ₁	0 ₃	0 ₂	
01	0 ₄	x ₅	x ₇	x ₆	
11	0 ₁₂	x ₁₃	x ₁₅	x ₁₄	
10	1 ₈	0 ₉	1 ₁₁	0 ₁₀	

D_1

xQ_2	Q_1Q_0	00	01	11	10
00	0	0	1	0	
01	1	x	x	x	
11	0	x	x	x	
10	0	1	0	1	

D_0

xQ_2	Q_1Q_0	00	01	11	10
00	1	0	0	1	
01	1	x	x	x	
11	0	x	x	x	
10	0	0	0	0	

$$D_2 = x(Q_1 + \bar{Q}_0)(\bar{Q}_1 + Q_0)$$

$$D_1 = \bar{x}Q_2 + x\bar{Q}_1Q_0 + \bar{x}Q_1Q_0 + xQ_1\bar{Q}_0$$

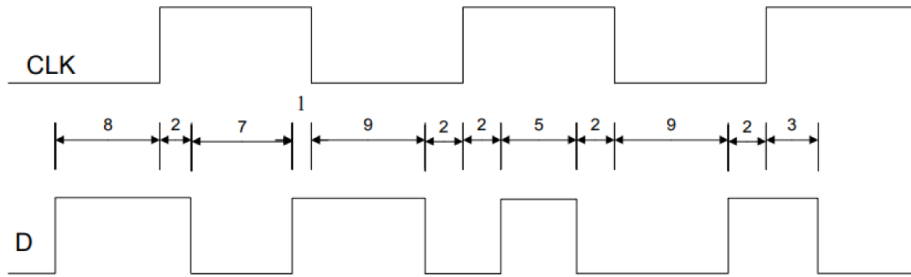
$$D_0 = \bar{x} \cdot \bar{Q}_0$$

y

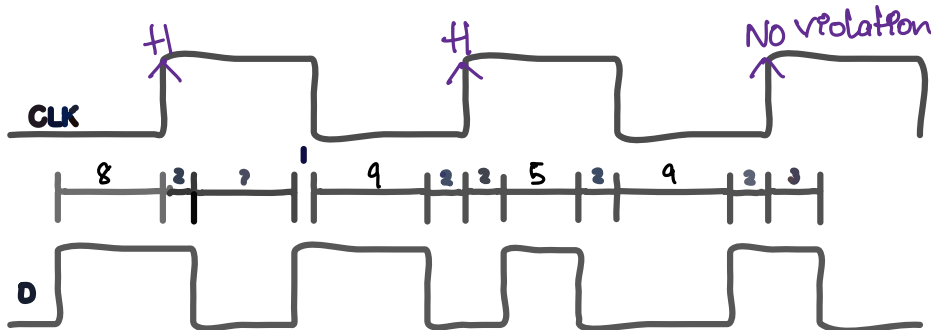
xQ_2	Q_1Q_0	00	01	11	10
00	0	1	0	0	
01	1	x	x	x	
11	0	x	x	x	
10	0	1	0	1	

$$y = \bar{x}Q_2 + \bar{Q}_1Q_0 + xQ_1\bar{Q}_0$$

- 14) A positive-edge triggered D-FF has a setup time of 2ns and a hold time of 3ns. If the waveforms shown below are applied to this FF, *circle* the timing violations and *label* them with the letters 'S' or 'H' depending on whether the violations are setup or hold time violations.



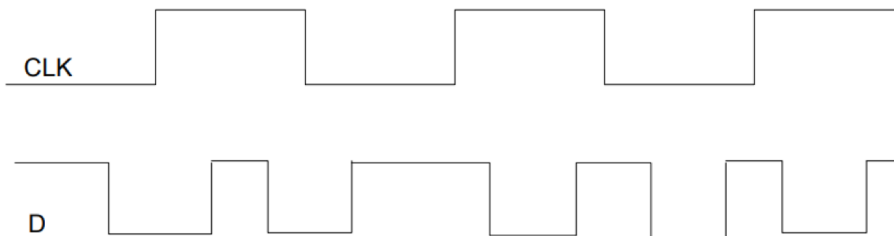
Soln:



$$T_{\text{setup}} = 2\text{ns}$$

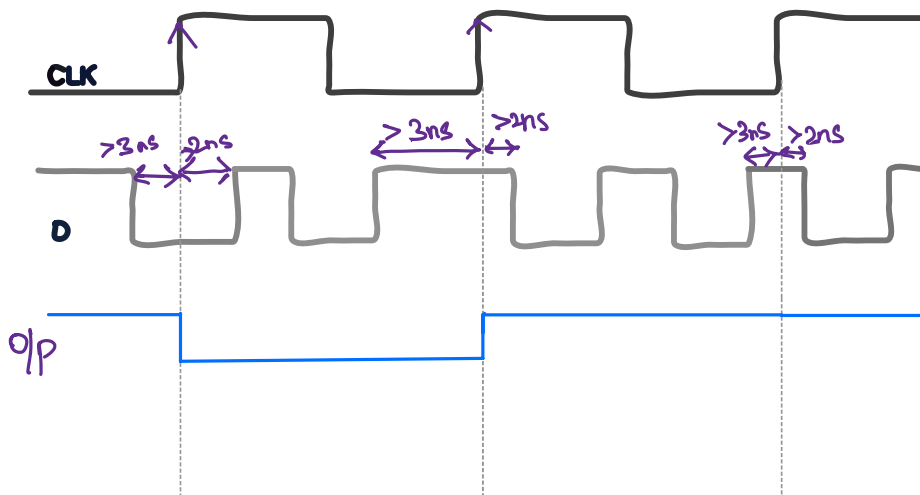
$$T_{\text{hold}} = 3\text{ns}$$

- 15) A level sensitive D-latch has a setup time of 3ns and a hold time of 2ns. If the waveforms shown below are applied to this FF, draw the output waveform(Q) and mark the timing constraints you would place on the D-input to satisfy the minimum setup & hold times. Assume that the initial FF-state is Q = 1.



$$T_{\text{setup}} = 3\text{ns}$$

$$T_{\text{hold}} = 2\text{ns}$$

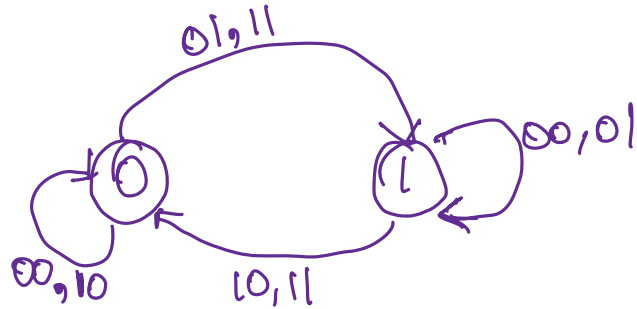


16) Draw the state diagram of the Moore machine which has 2 inputs (X_1, X_2) and one output Z . The output of the machine is determined by the following:

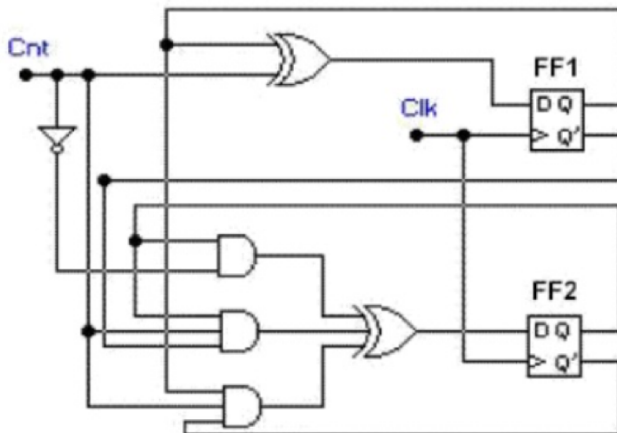
- Z does not change its value if $X_1X_2 = 00$
- Z becomes 1 if $X_1X_2 = 01$
- Z becomes 0 if $X_1X_2 = 10$
- Z changes its value if two consecutive 11 are received at the inputs X_1X_2
- The reset input initializes the machine to a 0 output

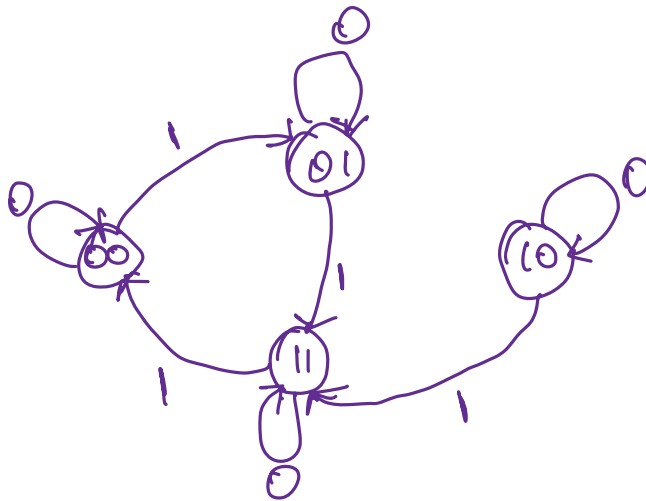
Soln:

X_1	X_2	Z	Z^+
0	0	0	0
0	1	0	1
1	0	0	0
1	1	0	1
0	0	1	1
0	1	1	1
1	0	1	0
1	1	1	0

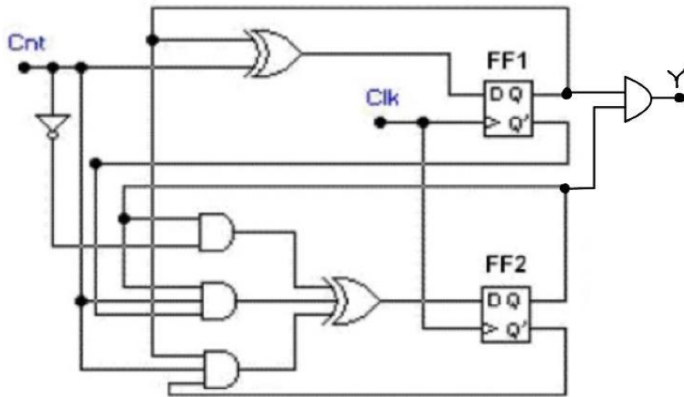


17) Derive the state table and state diagram for the synchronous circuit shown below:



$$\left\{ \begin{array}{cc} D_1 & D_2 \\ 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \end{array} \right\}$$


18) Draw the state diagram for the following circuit:



Soln

C	Q ₂	Q ₁	D ₂	D ₁	Q ₂ ⁺	Q ₁ ⁺	Y
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	1	0	1	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	0	1	0
1	0	1	1	0	1	0	0
1	1	0	1	1	1	1	1
1	1	1	0	0	0	0	0

$$Y = Q_1^+ Q_2^+$$

$$D_2 = Q_2 \bar{C} \oplus Q_2 C \bar{Q}_1 \oplus Q_1 C \bar{Q}_2$$

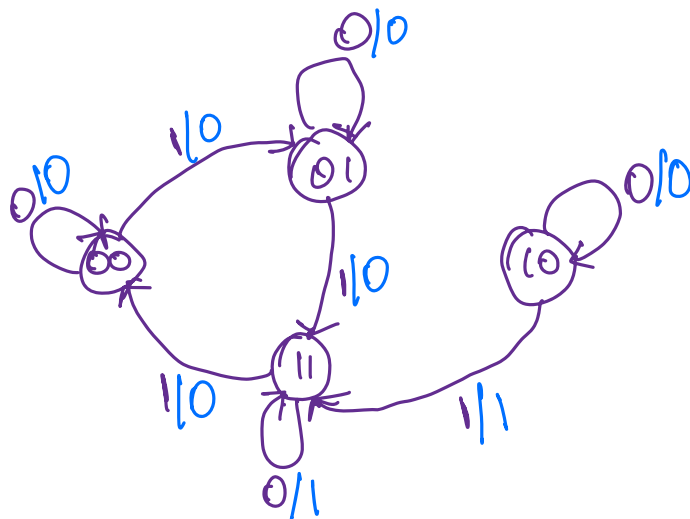
$$D_1 = Q_1 \oplus C$$

When $C=0$

$$D_2 = Q_2, D_1 = Q_1$$

When $C=1$

$$D_2 = \bar{Q}_1 Q_2 + \bar{Q}_2 Q_1, D_1 = \bar{Q}_1$$

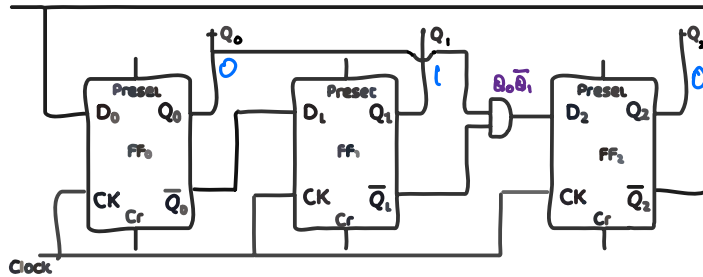
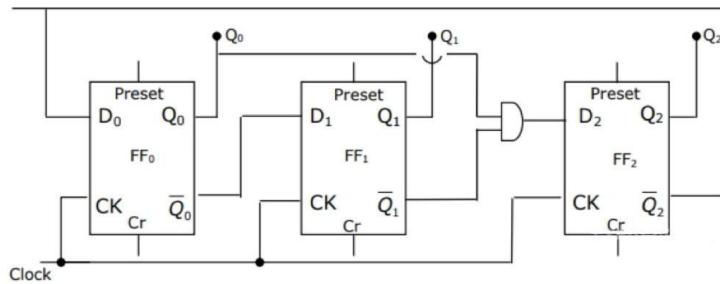


19) A sequence generator is shown in the figure. The counter status ($Q_0 Q_1 Q_2$) is initialised to 010 using preset/clear inputs.

The clock has a period of 50ns and transitions take place at the rising clock edge.

(a) Give the sequence generated at Q_0 till it repeats.

(b) What is the repetition rate for the generated sequence?



$T_{clk} = 50\text{ns}$
+ve edge

	$Q_0 Q_1 Q_2$	$\bar{Q}_2$	$\bar{Q}_0$	$Q_0 \bar{Q}_1$	$Q_0 + Q_1 + Q_2$
5	0 0 0	1	1	0	1 1 0
1	0 0 1	1	1	0	0 1 0
1	0 1 0	1	1	0	1 1 0
3	0 1 1	1	1	0	0 1 0
2	1 0 0	1	0	1	1 0 1
4	1 0 1	1	0	1	0 0 1
2	1 1 0	1	0	0	1 0 0
	1 1 1	0	0	0	0 0 0

010 $\rightarrow$ 110 $\rightarrow$ 100 $\rightarrow$ 101 $\rightarrow$ 001

20) Construct a state diagram for a synchronous sequential circuit that detects as input sequence of 1010. The output Z is 1 when the sequence is detected. Otherwise, Z is 0. Sequences are allowed to overlap. A sample input/output is given below.

Input X: 001011110100111010100100

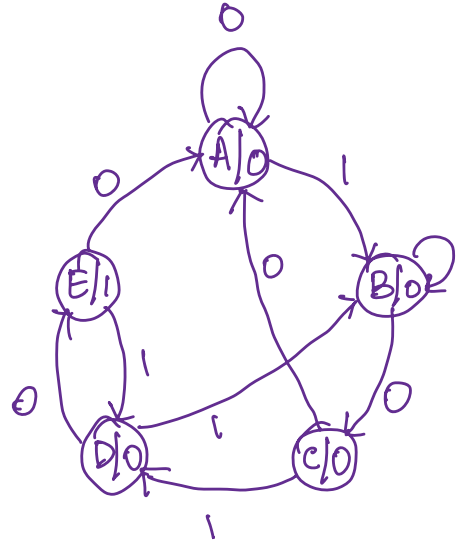
Output Z: 000000000001000000101000

Soln:

→

1
10
101
1010

A
B
C
D
E



22) An excitation table of a flip flop describes the inputs required to change state, depending on its current state. Excitation table for SR flip flop is provided below. Construct the excitation table of JK, D and T flip flops.

Truth table of JK flip flop.

J	K	Q_k	Q_{k+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Excitation table of JK FF

Q_k	Q_{k+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Truth table of D Flip Flop

D	Q_k	Q_{k+1}
0	0	0
0	1	0
1	0	1
1	1	1

Excitation Table of D FF

Q_k	Q_{k+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Truth table of T Flip Flop

T	Q_k	Q_{k+1}
0	0	0
0	1	1
1	0	1
1	1	0

Excitation Table of T FF

Q_k	Q_{k+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

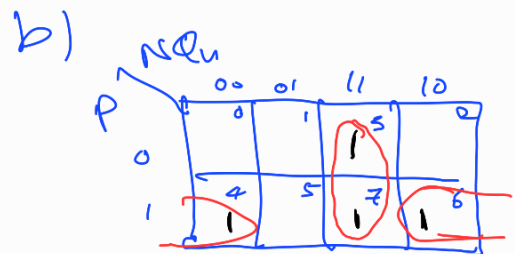
23) A PN flip-flop has four operations: clear to 0, no change, complement, and set to 1, when inputs P and N are 00, 01, 10, and 11, respectively.

- Tabulate the characteristic table.
- Derive the characteristic equation.
- Tabulate the excitation table.
- Show how the PN flip-flop can be converted to a D flip-flop.

Characteristic Table of PN FF

a)

P	N	Q_n	Q_{n+1}
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



$$Q_{n+1} = \overline{P}Q_n + NQ_n$$

c) Excitation Table

Q_n	Q_{n+1}	P	N
0	0	0	X
0	1	1	X
1	0	X	0
1	1	X	1

NC / reset

Toggle / set

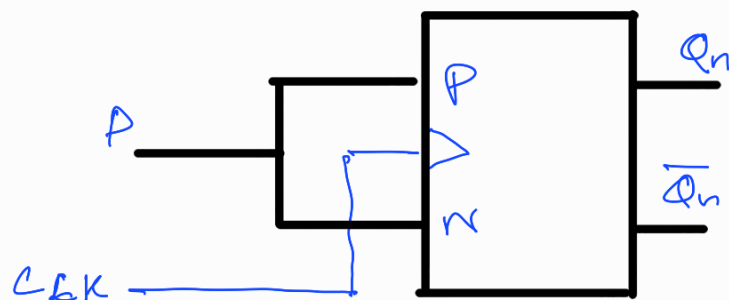
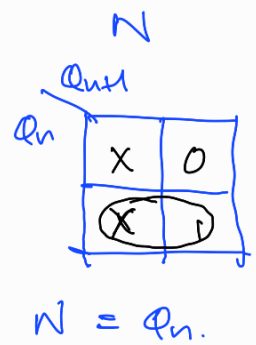
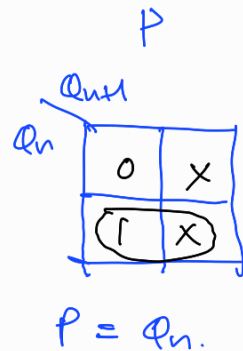
Toggle / reset

NC / set

d) D FF using PN FF

charac. of D:

D	Q_n	Q_{n+1}	P	N
0	0	0	0	x
0	1	0	x	0
1	0	1	1	x
1	1	1	x	1



24) Implement the following designs

- SR, JK and D Flip Flops using T flip flop

① choice of next FF

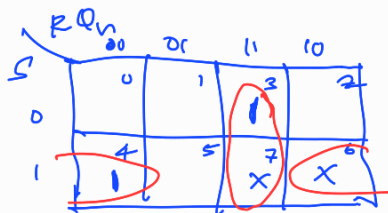
② Excitation table of available FF

SR Flip Flop using T flip Flop:

S	R	Q_n	Q_{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	X	X
1	1	1	X	X

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	X

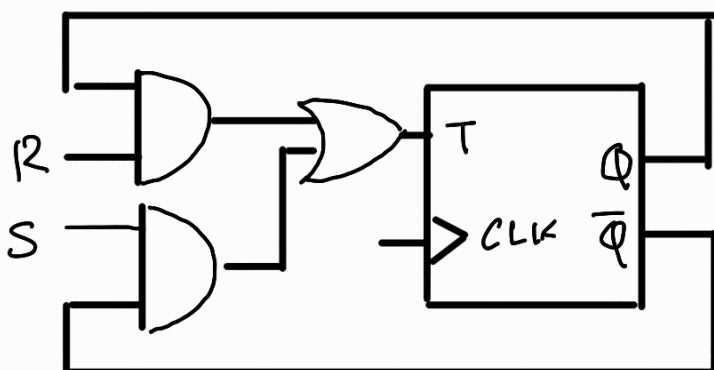
T	Q_{n+1}
0	Q_n
1	$\overline{Q_n}$



$$T = RQ_n + S\overline{Q_n}$$

Excitation table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

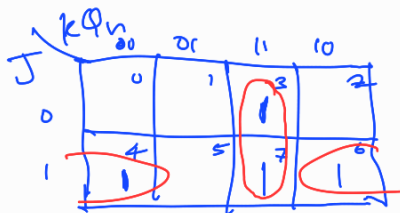


JK Flip Flop using T flip Flop:

J	K	Q_n	Q_{n+1}	T
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	1
1	0	1	1	0
1	1	0	1	1
1	1	1	0	1

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	X

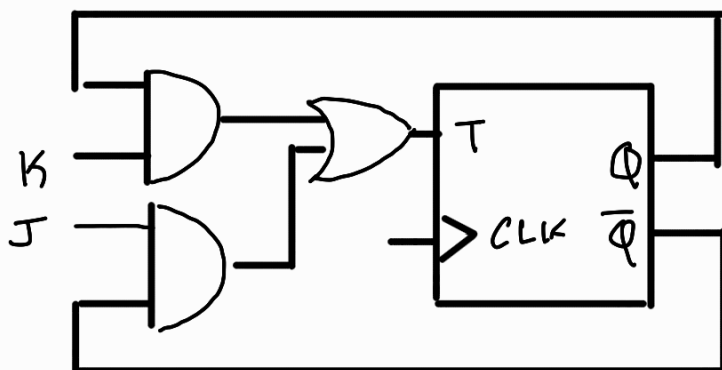
T	Q_{n+1}
0	Q_n
1	$\overline{Q_n}$



$$T = KQ_n + J\overline{Q_n}$$

Excitation table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0



D Flip Flop using T flip Flop:

D	Q_n	Q_{n+1}	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

$$T = \overline{D}Q_n + D\overline{Q}_n$$

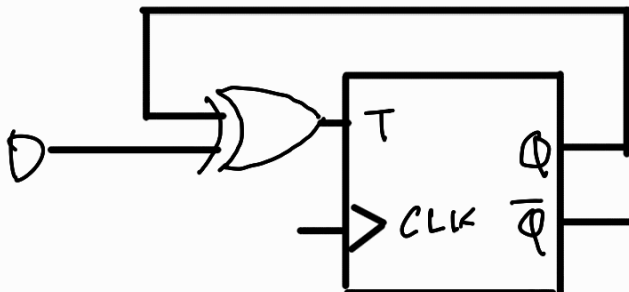
$$= D \oplus Q_n$$

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	X

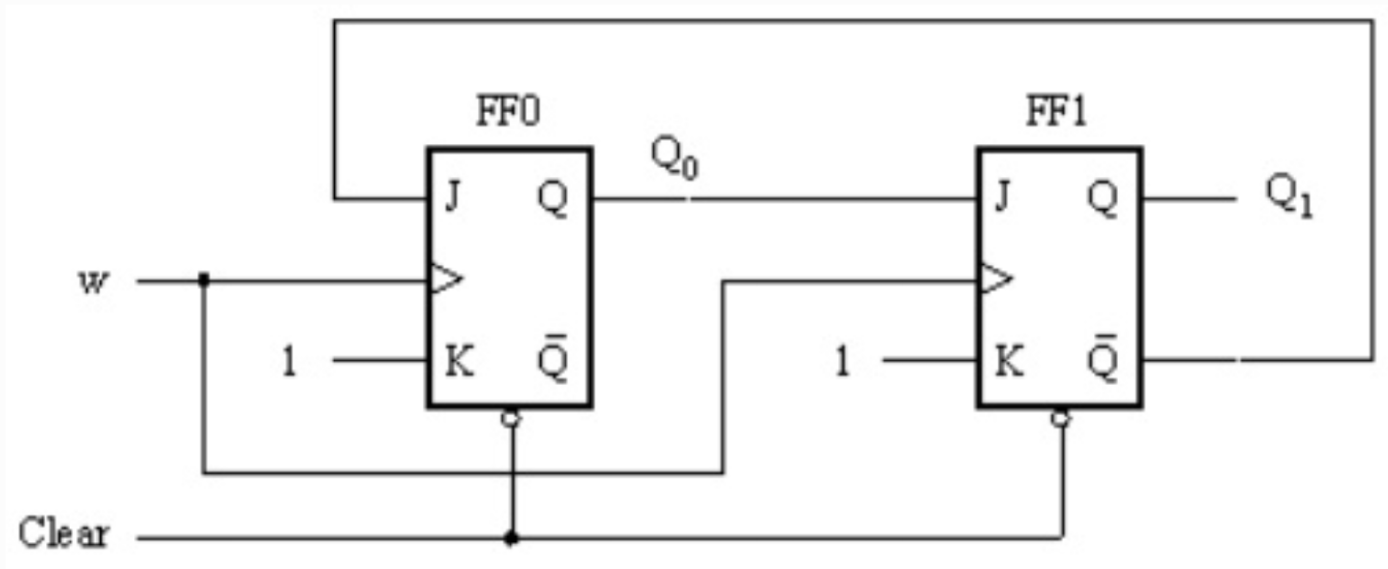
T	Q_{n+1}
0	Q_n
1	$\overline{Q}_n$

Excitation table

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0



25) Determine the functional behavior of the circuit in Figure. Assume that input w is driven by a square wave signal.



Step 1: find I/p & o/p Equations

$$\begin{aligned} J_0 &= \bar{Q}_1 & J_1 &= Q_0 & y &= Q_1 \\ K_0 &= 1 & K_1 &= 1 \end{aligned}$$

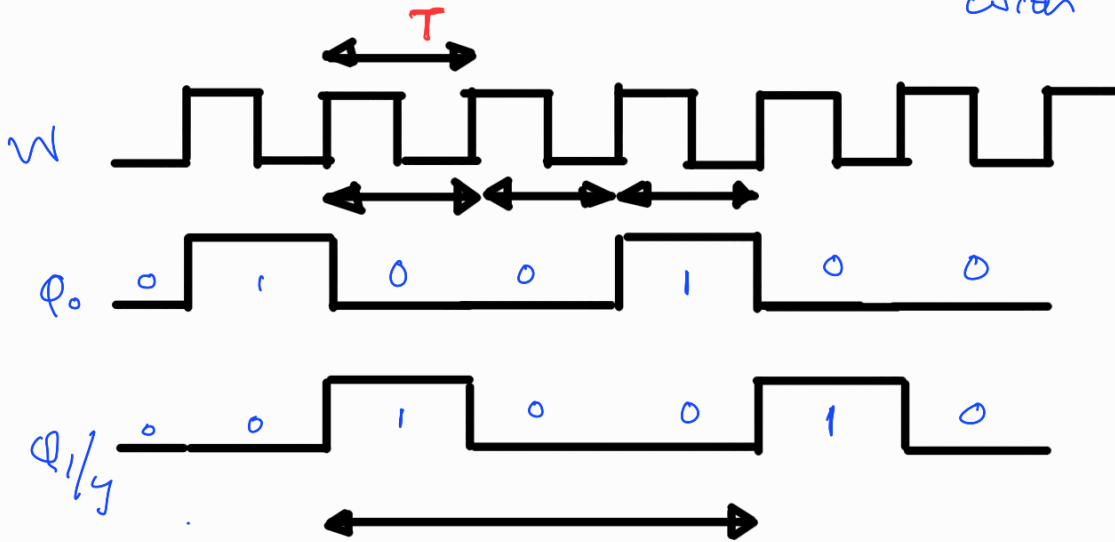
Step 2: Construct the state table.

Since K_0 and K_1 are set to 1, the excitation equation of JK FF becomes.

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n = J\bar{Q}_n$$

Present State		Inputs		Next State		Output
Q_0	Q_1	$J_0 = \bar{Q}_1$	$J_1 = Q_0$	$Q_0^+ = J_0 \bar{Q}_0$	$Q_1^+ = J_1 \bar{Q}_1$	$y = Q_1$
0	0	1	0	1	0	0
0	1	0	0	0	0	0
1	0	1	1	0	1	1
1	1	0	1	0	0	0

Step 3: Trace the Output: Consider Square Wave w with Time Period T



$$T_{out} = 3 \times T_{in}$$

$$f_{out} = \frac{f_{in}}{3}$$

$$\text{duty cycle} = \frac{100}{3}$$

Provided circuit divides the input frequency by 3.

26) A sequential circuit with 2 D flip-flops, A and B and an external input x, is specified by the following next-state equations:

$$A(t+1) = x'A + xB$$

$$B(t+1) = xA' + x'B$$

- List the state table showing present state and next states?
- Draw the corresponding state diagram?
- Explain the functionality of the circuit?

Step 1: write I/f & o/p Equations:

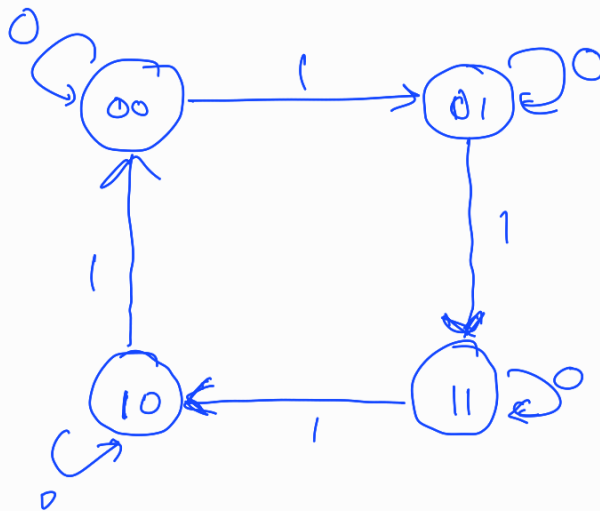
$$A^+ = \overline{x}A + xB.$$

$$B^+ = x\overline{A} + \overline{x}B.$$

Step 2: Construct state Table.

P.S		Input	N.S.		
A	B	x	$\overline{x}A + xB \Rightarrow A^+$	$x\overline{A} + \overline{x}B \Rightarrow B^+$	A ⁺ B ⁺
0	0	0	$1 \cdot 0 + 0 \cdot 0 = 0$	$0 + 0 = 0$	0 0
0	0	1	$0 + 0 = 0$	$1 + 0 = 1$	0 1
0	1	0	$0 + 0 = 0$	$0 + 1 = 1$	0 1
0	1	1	$0 + 1 = 1$	$1 + 0 = 1$	1 1
1	0	0	$1 + 0 = 1$	$0 + 0 = 0$	1 0
1	0	1	$0 + 0 = 0$	$0 + 0 = 0$	0 0
1	1	0	$1 + 0 = 1$	$0 + 1 = 1$	1 1
1	1	1	$0 + 1 = 1$	$0 + 0 = 0$	1 0

Step 2: State Diagram



(or 2 bit)

The described circuit is a Mod 4 up counter that used grey code for state assignment

27) Reduce the following state table:

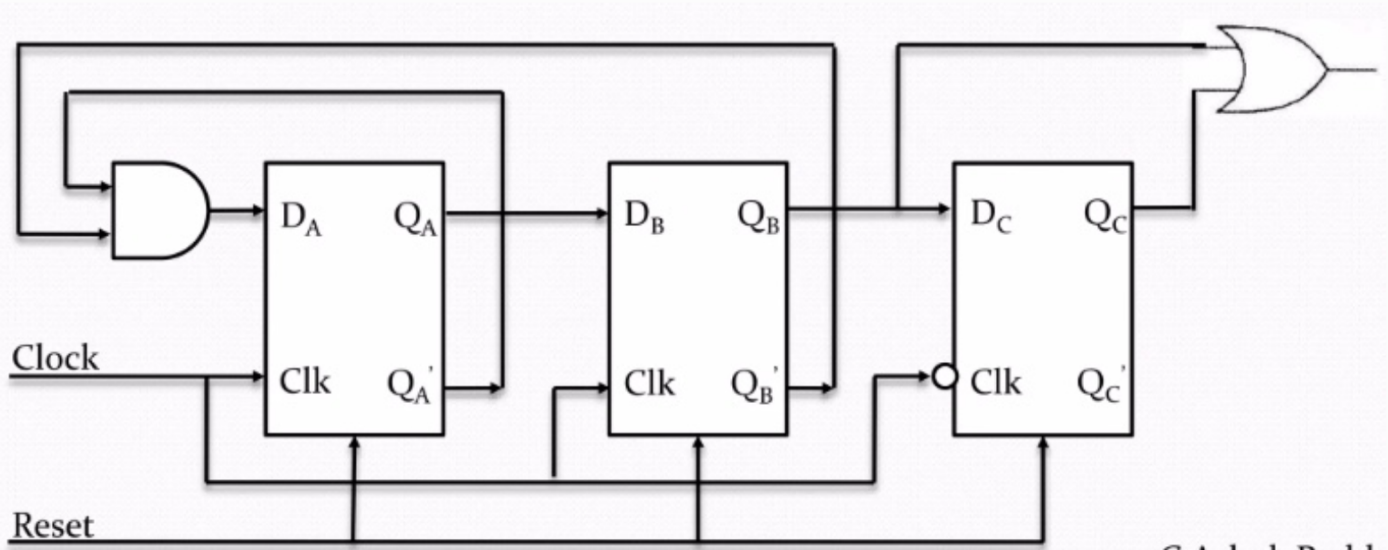
Present State		Next State		Output	
		x = 0	x = 1	x = 0	x = 1
1	a	a	b	0	0
	b	e	c	0	1
2	c	a	d	0	1
	d	e	f	0	1
	e	a	f	0	0
2	f = c	g = d		0	1
1	g = a	a		0	0

① $g = a$

② $f = c$

Two redundant states are removed to obtain the minimal possible state table.

28) Given a circuit as shown below, draw the timing diagram at all the nodes. Can you comment on what the circuit is doing to the input.



Step 1: Collect I/p & o/p equations:

$$D_A = \overline{Q_A} \cdot \overline{Q_B} \quad D_B = Q_A \quad D_C = Q_B \quad Y = Q_B + Q_C$$

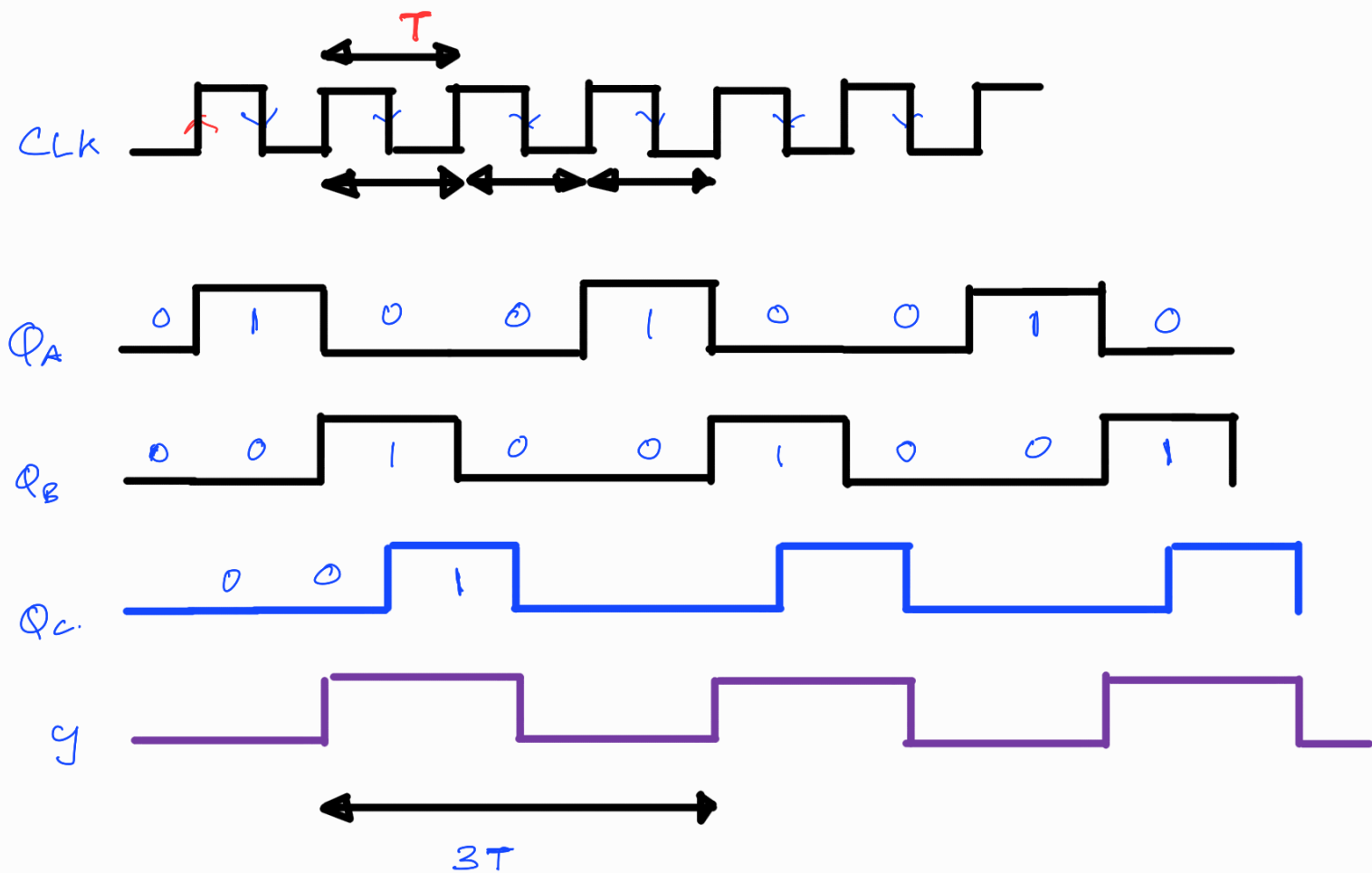
+ve edge of CLK
-ve edge of CLK

Step 2: Construct the state table.
Since D FF

P. S.			Inputs / Next states.		
Q_A	Q_B	Q_C	$D_A = \overline{Q_A} + \overline{Q_B}$	$D_B = Q_A$	$D_C = Q_B$
0	0	0	1	0	0
0	0	1	1	0	0
0	1	0	0	0	1
0	1	1	0	0	1
1	0	0	0	1	0
1	0	1	0	1	0
1	1	0	0	1	1
1	1	1	0	1	1

Out put
 $Y = Q_B + Q_C$

Step 2: Construct Outputs at each node



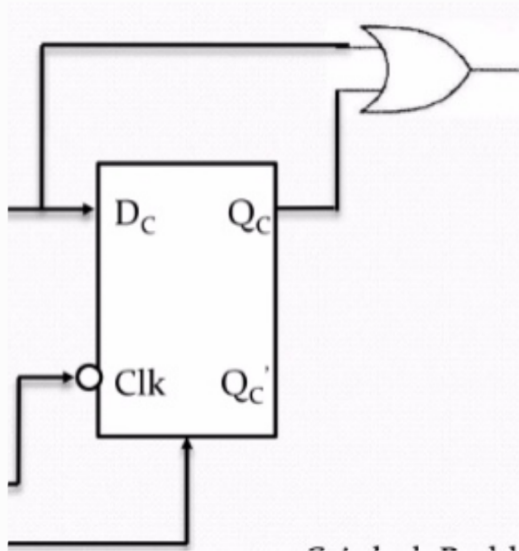
Time Period of output $T_{out} = 3 \times T_{in}$

$$\therefore f_{out} = \frac{f_{in}}{3} \quad \text{with 50\% duty cycle.}$$

Inference:

- ① Q_c changes its state during Negative edge of the clock.
- ② A +ve edge triggered FF can be used to delay a signal by 1 Clock Period
- ③ A negative edge triggered D FF can be used to delay a signal by a Half clock period

④



using such configuration
50% duty cycle is
obtained in a $f/3$
circuit